PAPER 2

QUESTION 1

(a) Describe the features of the secondary structure of **A1**. [2]

1. Comprises both α-helices and
2. β-pleated sheets
3. The α-helix takes the form of an extended spiral spring
4. with 3.6 amino acid residues per turn
5. The β-pleated sheet has an extended zigzag, sheet-like conformation
6. The α-helix and β-pleated sheet are stabilised by hydrogen bonds between C=O and NH groups of the A1 polypeptide backbone

(b) Explain how the structure of **A1** enables it to catalyse the hydrolysis of starch. [3]

1. A1 has an active site
2. with a specific 3D conformation that is complementary to that of starch substrate
3. so that starch binds to active site to form A1-starch complex
4. Ref to role of binding amino acid residues
5. Ref to role of catalytic residues, aspartic acid at positions 197 and 300 and glutamic acid at position 233
6. Critical α(1,4) glycosidic bonds are strained
7. thereby lowering activation energy of reaction
8. Ref to role of structural residues

(c) Account for the shape of the graph from **A** to **D**. [3]

A to B:
1. Increase in starch concentration increases frequency of effective collisions between A1 and starch molecules
2. hence increasing rate of A1-starch complex formation

B to C:
3. Extent of increase in A1 activity decreases as starch concentration increases as a smaller number of active sites is available at any one time

C to D:
4. Further increase in starch concentration does not result in increase in rate of reaction which is limited by saturation of A1 active sites
5. Starch concentration is no longer the limiting factor / A1 concentration is the limiting factor

(d) Suggest a reason for the shape of the graph from **D** to **E**. [1]

1. Ref to inhibition of A1
2. High starch concentration / end-product, changes conformation of A1 active site

(e) With reference to Fig. 1.1, suggest why **A2** has a higher optimum temperature than **A1**. [2]

1. More disulfide bonds – 9 in A2 compared to 5 in A1
2. Ref to covalent nature and strength of disulfide bonds
3. 3D structure of A2 is more stable than A1
4. A2 is more resistant to denaturation with increasing temperature

(f) Suggest the evolutionary significance of the existence of two forms of amylase **A1** and **A2** in *A.aquaticus*. [1]

A1 and A2 have different temp optima, so *A. aquaticus* can adapt to different environmental conditions to ensure survival

[Total: 12]

QUESTION 2

(a) Describe how the quaternary structure of adult haemoglobin is formed. [2]

1. Two α and 2 β globin chain/ subunit form the tetrameric protein
2. held together by weak hydrophobic interactions, ionic and hydrogen bonds
3. to form a compact and spheroidal structure

(b) Explain how β-LCR influences expression of β gene in adults. [3]

1. β-LCR is an enhancer sequence
2. where activator proteins bind to
3. to increase / maximise the transcription of β gene
4. ref. to recruitment of RNA polymerase and GTFs to form the transcription initiation complex

(c) Suggest and explain how chromatin remodelling can prevent transcription factors from accessing the δ gene. [3]

1. DNA methylation is the covalent modification by addition of methyl groups
2. to the promoter of a gene / cytosine nucleotides/ CpG dinucleotides after DNA replication
3. catalysed by DNA methyltransferases
4. changes the 3D conformation of DNA
5. and prevents the binding of transcription factors to the promoter which is required for transcription initiation.

or

1. Deacetylation of acetylated lysine residues in histone tails
2. catalysed by histone deacetylases (HDACs)
3. lysine residues become positively charged
4. results in increased affinity of the histone complex for the DNA molecule.
5. hence, the chromatin structure becomes more compact / prevents access of transcription factors to the promoter which is required for transcription initiation.

(d) Apart from chromatin remodelling,
(i) suggest how CRC proteins regulate the expression of β gene. [2]

1. Ref. to the identify of CRC protein as a bending protein
2. Idea of causing the DNA to bend / changing conformation of DNA
3. hence bringing β-LCR to, closer proximity / AW, to β gene
4. Idea of CRC protein having dual role of activating β gene and repressing ε, Gγ, Aγ

(ii) describe two differences between prokaryotic and eukaryotic gene regulation at the transcriptional level. [2]

Eukaryotic Gene Expression	Prokaryotic Gene Expression
1. Transcription of each gene is controlled individually by its own promoter 2. Promoter proximal elements are located close to promoter while distal control elements are far away from the promoter. 3. Distal control elements control transcription 4. Combinational control	1. Several functionally related genes are regulated by the same promoter to form an operon 2. Control elements are found near or at the promoter. 3. Control elements near or at the promoter regulate transcription 4. No combinational control possible.

[Total: 12]

QUESTION 3

(a) (i) With reference to the *lac* operon, describe the concept of a simple operon. [2]

1. A simple operon, such as the lac operon, consists of functionally related structural genes, *lacZ*, *lacY* and *lacA*, which codes for enzymes that are involved in lactose metabolism
2. These structural genes are placed under the control of the same promoter, operator and regulatory gene, *lacI*, for RNA polymerase to bind and initiate transcription
3. Ref to the concept of a simple operon without ref to the lac operon

(ii) Suggest the effect on the transcription of the *lacZ* gene when lactose is absent. [2]

1. In the absence of lactose, lac repressor protein binds to the operator region, which is now upstream of the promoter / ref to the effect of the inversion
2. Hence, RNA polymerase is not prevented from binding to the promoter, transcription of *lacZ* gene is now possible

(b) (i) Explain the increase in the β-galactosidase mRNA level in culture flask **A** after one dose of lactose was added into the culture medium. [3]

Without glucose, bacteria now utilize lactose:

1. Allolactose, an isomer of lactose acts as an inducer and binds to the lac repressor
2. Lac repressor assumes an inactive conformation and cannot bind to the operator
3. RNA polymerase can now bind to the promoter
4. to allow for transcription of *lacZ* gene

5. In the absence of glucose, cAMP concentration increases
6. and binds to the catabolite activator protein (CAP)
7. CAP assumes an active conformation to binds to the CAP binding site;
8. To enhance the binding of RNA polymerase to the promoter, upregulating transcription

(ii) In Fig. 3.1, sketch a graph to indicate the β-galactosidase mRNA level in culture flask B before and after the addition of IPTG. Label this graph as **B**. [1]

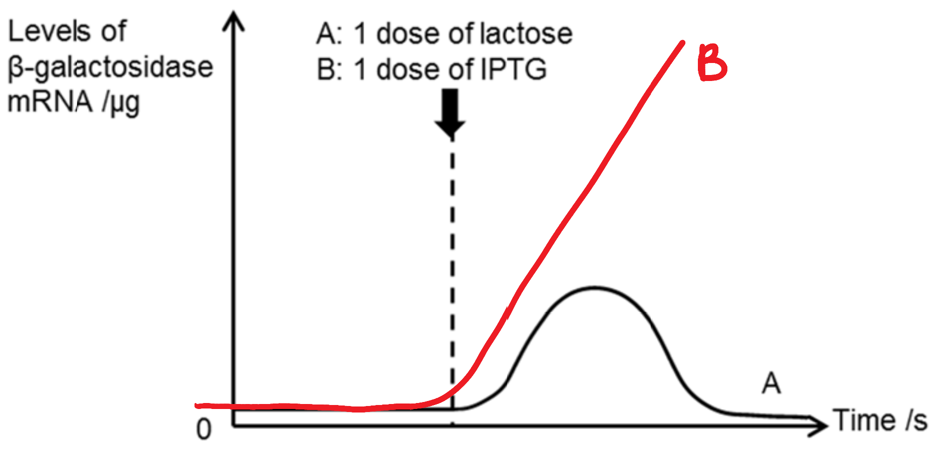

Fig. 3.1

(iii) Account for the sketch in (b)(ii). [3]

<u>Before dosage:</u>

1. mRNA levels are as low as A
2. With only glucose present, β-galactosidase is not required to hydrolyse lactose/ AW to account for low mRNA levels

<u>After dosage:</u>

3. IPTG acts as an inducer by binding to the lac repressor
4. Lac repressor switch to inactive conformation and does not bind to the operator region
5. RNA polymerase can now bind to the promoter
6. allow transcription of *lacZ* gene
7. As IPTG cannot be degraded by β-galactosidase, it remains bound to the lac repressor
8. promoting continuous transcription of the lac operon, resulting in the increasing levels of β-galactosidase mRNA

[Total: 11]

QUESTION 4

(a) Define the term *pure-breeding*. [1]

A homozygous individual / organism with two identical alleles in one gene

(b) (i) Draw a genetic diagram to show the results of the test cross. [5]

phenotypes	non-banded with crossveins	X	banded without crossveins
genotypes	**<u>bD</u>** / **Bd**	X	**<u>bd</u>** / **bd**
Gametes	**<u>bD</u>** **<u>Bd</u>** **<u>bd</u>** **<u>BD</u>**		**<u>bd</u>** ;

		female Gametes			
		<u>bD</u>	**<u>Bd</u>**	**<u>bd</u>**	**<u>BD</u>**
male Gamete	**<u>bd</u>**	**<u>bD</u>** / **bd**	**<u>Bd</u>** / **Bd**	**<u>bd</u>** / **bd**	**<u>BD</u>** / **bd**

phenotype	**banded with crossveins**	**non-banded without crossveins**	**banded without crossveins**	**non-banded with crossveins**
observed number	483	512	21	35
	parental		recombinant	

(ii) Using information provided on Drosophila, explain how genotype is linked to phenotype. [3]

1. A genotype is the genetic makeup of an organism / results from a combination of alleles
2. It refers to the paired alleles that produces a phenotype / physical character / physical manifestation; In Drosophila, when the genotype of the organism consists of at least one copy of a dominant allele of the gene for band
3. Drosophila do not display a dark transverse band in the thorax
4. If the genotype consists of a lack of dominant allele / two copies of recessive alleles of the gene for presence of dark transverse band

(iii) If the two genes on *Drosophila* are found very close together on the same chromosome, explain how the test cross results would differ. [2]

1. Only 2 phenotypes will be seen / no recombinants will be seen / phenotypic ratio of 1:1 will be observed
2. As the two genes in Drosophila will undergo complete linkage / inherited together as one unit
3. No crossing over will occur to produce recombinants

[Total: 11 marks]

QUESTION 5

(a) (i) Identify structures **X** and **Y**. [2]

X : intergranal lamella Y : stroma

(ii) Explain how the **Z** is adapted to carry out photophosphorylation. [2]

1. Z, is made from a stack of thylakoid membranes, / provides a large surface area
2. Allows for embedding of many, photosynthetic pigments / photosystems / light harvesting complexes, for light, harvesting / absorption
3. The thylakoid membrane is impermeable to H^+ thus protons can only diffuse through stalked particles embedded in the membrane down its concentration gradient

(b) (i) Describe and explain the effects an increase in oxygen concentration will have on photosynthesis. [3]

1. An increase in oxygen concentration reduces the rate of photosynthesis / increases the rate of photorespiration
2. This is less Rubisco is available for CO_2 / more oxygen competing with CO_2 for Rubisco
3. This results in less CO_2 , fixation for Calvin cycle
4. There is less, glycerate 3-phosphate / GP / TP , produced and less RuBP being, regenerated

(ii) Suggest why the process outlined in Fig 5.2 is known as photorespiration. [2]

1. The process uses oxygen and, excretes / produces , carbon dioxide;;
2. Light energy / non-cyclic photophosphorylation / light dependent reaction / products of the light dependent reaction / ATP and NADPH, is required for this process to occur
3. The same photosynthetic enzyme / Rubisco is used and allows the Calvin cycle to continue

(c) Suggest why photorespiration does not occur in these plants. [1]

oxygen's 3D conformation is not complementary to PEP carboxylase, thus oxygen is not a substrate for / cannot bind to / will not compete for PEP carboxylase

[Total: 10]

QUESTION 6

(a) Explain why cells in multicellular organism need to communicate with one another. [2]

1. The different cells need to collaborate to coordinate /regulate their activities such as growth, differentiation, control of movement/AW
2. Cells need to receive and respond to information from the internal and external environment in an integrated manner/AW
3. Such that the organism develops and functions as an integrated and coherent whole/AW
4. Communication is required for cells to interact to mediate the maintenance of homeostatic bodily processes/AW

(b) (i) State the name given to the gap between the two neurons at this junction. [1]

Synaptic cleft

(ii) With reference to the fluid mosaic model, compare the plasma membrane of the first and second neurons. [2]

Similarities:

1. Plasma membrane of both neurons are made of phospholipid bilayer and with proteins that are embedded in the phospholipid bilayer
2. Phospholipid and proteins are able to move laterally in both membranes

Differences:

1. Serotonin receptors are present on the second neuron but they are absent from the first neuron
2. The first neuron has voltage-gated Ca^{2+} channel, while the second neuron has chemically-gated Na^{+} channel

(iii) Outline the significance of junctions between neurons in the function of the nervous system. [2]

1. It allows for transmission of nerve impulse in the form of neurotransmitters / chemical impulse from presynaptic neuron to the postsynaptic neuron
2. via the diffusion of neurotransmitters across the synaptic cleft
3. One neurone can, receive impulse from / transmit impulses to, many neurons
4. It allows for summation of postsynaptic membrane potential
5. To ensure that only stimulation that causes depolarization above the threshold will generate action potential in the postsynaptic neurone
6. It also ensures that the movement of nerve impulse occurs only in one direction / unidirectional transmission

(c) (i) Describe the effect of MDMA on the movement of the three groups of mice. [3]

1. Before they were given MDMA, all groups of mice showed little movement, with path length of less than 50cm from 0-5 minutes
2. Upon administration of MDMA at the 5th minute, group L shows increasing path length from 40-50cm at min 5 to ,470-495cm / almost 500cm, after 10 minutes
3. However, group K & M does not show significant change in the amount of movement
4. As the path length remains relatively the same at 40-80cm throughout after 10 minutes

(ii) Suggest how MDMA affects movement based on the trend observed in Fig 6.2. [2]

1. MDMA binds to serotonin receptor / requires serotonin receptor (to affect movement);;
2. as MDMA has similar shape as serotonin / MDMA is complementary in shape to serotonin receptor;
3. causing depolarization of postsynaptic membrane / generating EPSP;
4. When action potential is generated (in muscle cells);
5. It will result in muscle contraction;
6. hence increased movement of mice;

[Total: 12]

QUESTION 7

(a) (i) Explain why it is biologically important that mtDNA includes genes for cytochrome c. [2]

1 Mitochondria is, the site of aerobic respiration / where ATP is produced
2 Cytochrome c is an electron carrier and is part of electron transport chain
3 genes not in nucleus
4 ref to some advantage of having genes in mitochondria e.g. trafficking from cytosol to mitochondria not necessary
5 AVP

(ii) Suggest how all the mitochondria in a male muscle cell derive from the mitochondria of his mother. [1]

1 new mitochondria form by, division / binary fission, (of pre-existing) mitochondria
2 mitochondrion originates in (cytoplasm of), ovum / oocyte (of mother)
3 no mitochondria from father pass into the ovum through the sperm / paternal mitochondria degenerate
4 muscle cell arises from (successive) divisions of (single) zygote
5 AVP

(iii) Suggest how a group of genes would be transmitted only to male offspring. [1]

Y chromosome

(b) (i) Deduce the conclusions that can be drawn from Fig. 7.2. [3]

1 the differences between the DNA of modern humans are much less than the differences between modern humans and Neanderthals
2 data quote: (members of the sample group of) modern humans showed approx / on average, 7 / between 6 and 8
3 the sample of Neanderthal DNA lay outside this range / Neanderthal DNA is distinctly different to that of modern humans
4 data quote: Neanderthal DNA had an average of 26 / 25 to 27 differences to DNA of modern humans
5 humans and Neanderthals are more closely related to one another than to chimpanzees
6 Neanderthals are evolutionarily closer to humans than to chimpanzees

(ii) Explain what is meant by this statement '*changes in mtDNA were relatively neutral in terms of evolution*'. [2]

1 unlikely to have an effect on the person's chance of survival
2 therefore unlikely to be influenced by natural selection
3 change in triplet without changing amino acid / change in amino acid without any change to the protein structure / function
4 hence no effect on the phenotype of the organism
5 AVP

(c) Suggest what might have been the circumstances which led to one woman giving rise to the whole modern human race. [3]

1 original population very small / population fell to a low level / reference to, genetic bottleneck / genetic drift
2 natural selection intense / chances of survival low / high child mortality rates
3 Eve / Eve's clan (might have) had adaptations that made them more likely to survive than others / survival of the fittest / differential reproduction
4 Eve's offspring survived by chance when the other humans died / correct reference to founder effect
5 Eve's offspring out-competed existing types of human
6 AVP

[Total: 12]

QUESTION 8

(a) Outline the significance of each stages of mitosis [6]

Prophase
1. Chromatin condenses to form chromosome
2. To prevent entanglement / breakage of chromatin
3. Each chromosome is made of two identical sister chromatids held by centromere

Metaphase
4. Each homologous chromosome line up singly on the equatorial plate
5. Kinetochore spindle fibres attach to kinetochore at centromere
6. No pairing of homologous chromosomes / no independent assortment between homologous chromosomes

Anaphase
7. Kinetochore spindle fibres shorten / contracts
8. Centromere segregate into two
9. Two sister chromatids separate and moves to opposite poles

Telophase
10. Nuclear envelope reforms
11. Thus protecting DNA from being degraded
12. Chromosome uncoil and decondense into chromatin
13. So that the cell is transcriptionally active again

Overall significance
S1. To ensure genetic stability / integrity
S2. Two identical sister chromatids are distributed equally / same number of chromosomes in the two daughter cells / complete diploid set of chromosomes
S3. Therefore, no genetic variation in the daughter cells / genetically identical daughter cells

(b) With reference to one named example each, explain the genetic basis of continuous variation and discontinuous variation. [8]

Continuous variation

C1. Named examples: height and weight of man
C2. A characteristic is controlled by the combined effect of multiple additive genes / polygenic inheritance
C3. Phenotypes not clear cut / a range of phenotypes
C4. Intermediates are observed
C5. Phenotypes can be modified by the cumulative effect of varying environmental factors acting on the different genotype
C6. Quantitative inheritance
C7. as statistical analyses give estimates of population parameters such as mean and standard deviation

Discontinuous variation

1. Named examples: pea flower colour, pea plant height, ABO blood group
2. A characteristic is controlled by one or a few genes
3. This gene may have two / more alleles
4. Phenotypes are definite and clear cut / they can be divided discrete phenotypic classes
5. Intermediates are not observed
6. There is little / no environmental effect on the phenotypic expression of the genes
7. Qualitative inheritance
8. Analysis done by making counts and ratios

(c) Describe the causes of genetic variation in a population. [6]

1. During prophase I of meiosis, crossing-over takes place between non-sister chromatids
2. resulting in recombination of segments of non-sister chromatids between homologous chromosomes
3. This leads to the formation of new combinations of alleles in gametes
4. At metaphase I, independent assortment of homologous chromosomes
5. The orientation of the bivalents with respect to the poles is random
6. Hence a 50% chance that a daughter cell gets a paternal chromosome or a maternal chromosome
7. Random fusion / fertilisation of gametes during sexual reproduction
8. carrying different combinations of chromosomes adds to genetic variation of the zygote formed
9. Mutations occurs to generate new alleles

QUESTION 9

(a) Using two named examples, outline how aberrations to the number of chromosomes occur and their effects on the phenotype of the organism. [6]

1. In a disease known as 45, X Turner Syndrome;
2. Loss of a single chromosome
3. results in monosomy
4. Monosomy in humans only occur for X chromosome
5. Affected females have only 45 chromosomes/ just a single X chromosome
6. This is due to a failure of chromosomes or chromatids
7. to separate and move to opposite poles
8. during anaphase I and anaphase II of meiosis; respectively.
9. In a disease known as 47, XXY Klinefelter syndrome
10. A gain of one chromosome; in a gamete
11. results in trisomy
12. whereby fertilising this gamete with a normal haploid gamete produces a zygote with three copies of the sex chromosomes

13. Affected males have more than 1 X chromosome;
14. This is due a failure of chromosomes or chromatids;
15. to separate and move to opposite poles;.
16. during anaphase I and anaphase II of meiosis; respectively.
17. In Down Syndrome;;,
18. A gain of one chromosome; in a gamete
19. results in autosomal trisomy;
20. whereby fertilising this gamete with a normal haploid gamete produces a zygote with three copies of chromosome 21
21. Affected individuals have an extra chromosome 21
22. This is due to a failure of chromosomes or chromatids
23. to separate and move to opposite poles
24. during anaphase I and anaphase II of meiosis; respectively.

(b) With reference to cystic fibrosis, explain how a change in the sequence of the DNA nucleotide may affect the amino acid sequence in a protein, and hence the phenotype of the organism. [8]

1. A deletion of the triplet base TTT in the template strand of the cystic fibrosis transmembrane regulator (CFTR) gene
2. leading to a change in the mRNA sequence of the CFTR during transcription
3. Hence, the amino acid / primary sequence of the protein is altered since there is an absence of the 508^{th}amino acid residue, phenylalanine
4. This causes a change in the specific three-dimensional conformation;; of the CFTR protein.
5. This results in the synthesis of a non-functional CFTR channel
6. A non-functional CFTR channel does not allow Cl^- to diffuse out of epithelial cells
7. due to an absence/ a reduction in the steepness of water potential gradient
8. Water cannot leave the cell via osmosis
9. This in turn results in the production of thick mucus
10. narrowing of air passages / reduces air flow and causes breathing to be more difficult
11. increased diffusion distance for gas exchange at the lungs, leading to insufficient oxygen being supplied to CF patient
12. bacterial infection;;, as the thick mucus serves as a breeding ground for bacteria.
13. inability of cilia to sweep away bacteria, fungi spores and other particles that we may take in with every breath
14. blockage of ducts / blockage of pancreatic ducts which results in poor digestion / blockage of ducts of the reproductive organs, which results in fertility problems / sterility
15. Serious infections as well as liver / pancreatic / digestive failures
16. cause malnutrition and poor growth

(c) Describe the development of cancer as a multi-step process. [6]

1 It is a multistep process due to accumulation of independent mutations;
2 in cancer-critical genes;
3 occurring in the lineage of a single cell;.
4 There are gain-of-function mutations in proto-oncogenes;
5 and loss-of-function mutation in tumour suppressor genes;
6 This mutations result in uncontrolled cell division;
7 whereby the rate of cell division is greater than the rate of cell death;
8 Activation of telomerases; occurs whereby
9 telomerase gene is expressed to produce telomerase;
10 to lengthen telomere/maintain telomere length;
11 so cancerous cells can continue to divide infinitely/ reach replicative immortality /escape Hayflick limit/escape replicative senescence (any one);
12 Angiogenesis is the formation of new blood vessels;
13 Genes coding for angiogenesis-activating proteins are expressed in cancer cells;
14 to produce angiogenesis-activating proteins to form blood vessels;
15 These blood vessels deliver nutrients and oxygen, and remove waste products; from tumour
16 Metastasis is a process of direct migration and penetration / invasion of cancer cells into surrounding tissues;
17 allowing cancer cells to be transported by the circulatory system/bloodstream;
18 to establish new secondary tumours; at these sites.

PAPER 1

1	B	6	C	11	A	16	C	21	D	26	D	31	B	36	C
2	B	7	B	12	A	17	C	22	C	27	A	32	D	37	B
3	A	8	D	13	A	18	D	23	A	28	D	33	C	38	B
4	C	9	D	14	B	19	D	24	A	29	D	34	D	39	B
5	D	10	A	15	C	20	B	25	A	30	C	35	A	40	C